

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

ILLUMINA INC.,
Petitioner,

v.

MOLECULAR LOOP INNOVATIONS LLC,¹
Patent Owner.

IPR2024-00964
Patent 11,041,852 B2

Before ANNETTE R. REIMERS, SUSAN L. C. MITCHELL, and
ZHENYU YANG, *Administrative Patent Judges*.

PER CURIAM.

JUDGMENT

Final Written Decision
Determining No Challenged Claims Unpatentable
35 U.S.C. § 318(a)

¹ After institution, Patent Owner changed its name from Molecular Loop Biosciences, Inc. to MolecularLoop Innovations LLC. *See* Paper 11, 1. We modify the caption accordingly.

I. INTRODUCTION

Illumina, Inc. (“Petitioner”) filed a Petition (Paper 1, “Pet.”) seeking *inter partes* review of claims 1–8 of U.S. Patent No. 11,041,852 B2 (Ex. 1001, “the ’852 patent”). We instituted trial to review the challenged claims. Paper 7 (“DI”).

Thereafter, Molecular Loop Innovations LLC (“Patent Owner”) filed a Response to the Petition (Paper 18, “PO Resp.”), Petitioner filed a Reply to Patent Owner’s Response (Paper 25, “Reply”), and Patent Owner filed a Sur-reply to Petitioner’s Reply (Paper 33, “Sur-reply”).

An oral hearing for this proceeding was held on September 23, 2025, and the transcript of that hearing is of record. *See* Paper 39 (“Tr.”).

The Board has jurisdiction under 35 U.S.C. § 6 and issues this Final Written Decision pursuant to 35 U.S.C. § 318 and 37 C.F.R. § 42.73. For the reasons provided below, we find Petitioner has not shown, by a preponderance of the evidence, the unpatentability of any of the challenged claims.

A. *Real Parties in Interest*

Each party identifies itself as the real party in interest. Pet. 64; Paper 11, 1.

B. *Related Matters*

According to the parties, the ’852 patent is involved in the following cases: *Molecular Loop Biosciences, Inc. v. Illumina, Inc.*, No. 1:24-cv-00680 (D. Del.) and *Illumina, Inc. et al. v. Molecular Loop Biosciences, Inc.*, No. 1:24-cv-11501 (D. Mass.). Pet. 64–65; Paper 3, 1. Patent Owner later informed us that the case before the District Court of Massachusetts is no longer pending. Paper 11, 1.

Petitioner also filed IPR2024-00965, challenging the claims of U.S. Patent No. 11,768,200, a patent in the same family as the '852 patent. Pet. 65.

C. The '852 Patent

The '852 patent relates generally to “methods for maintaining the integrity and identification of a nucleic acid template in a multiplex sequencing reaction.” Ex. 1001, Abstract. The methods include the steps of “obtaining a template nucleic acid, incorporating a pair of sequence identifiers into the template, and sequencing the template.” *Id.*

The '852 patent states that “[n]ext generation sequencing [“NGS”] typically involves the generation of an in vitro library; arraying of nucleic acid templates into physically distinct locations on a solid support (and optional clonal amplification of the templates into spatially localized clusters); and the sequencing-by-synthesis reaction itself.” *Id.* at 1:43–48.

According to the '852 patent, maintaining accurate sample identification during a multiplex sequencing reaction is an issue because “the sample preparation phase frequently introduces errors into the subsequent sequencing reaction.” *Id.* at 1:37–39. “For example, template molecules may physically overlap on a substrate; and in some cases result in a detectable erroneous signal.” *Id.* at 1:39–41; *see also id.* at 1:52–54 (stating that errors can be introduced from “improper pairing of template strands from different samples,” which can “form during the amplification of the multiplexed samples”). Because these errors have a significant effect on the results, the '852 patent states that “methods for reducing or eliminating errors in such procedures are needed.” *Id.* at 1:60–62.

The '852 patent explains the undesired introduction of sequencing errors when using a single barcode identifier. *See id.* at 4:17–63, Fig. 1 (showing “one example of how molecular overlap can lead to apparent high quality, but incorrect sequence data”). The '852 patent states that its method uses “two or more identifiers that are uniquely associated with an analyte of interest, such that the identifiers will only be present when the analyte of interest is detected.” *Id.* at 2:5–8. According to the '852 patent, “[i]n this way, false positive and/or false negative results are avoided by requiring concomitant detection of two or more independent markers associated with the analyte to be detected.” *Id.* at 2:8–11.

The '852 patent explains that

in sequencing applications, a pair of barcodes are placed such that they flank the sequence of interest. The identifiers are unique to the sample, so that valid sequence data are confirmed by the presence of the pair of flanking identifiers. In contrast, template containing only one identifier or an incorrect pair (i.e., a pair or more of identifiers not associated with the same template) will be excluded from analysis. Thus, methods of the invention make it possible to determine whether the integrity of a sample was maintained during a multiplex sequencing reaction, and prevent the assigning of sequence data to the wrong sample.

Id. at 2:18–29; *see also id.* at Fig. 2, 4:64–6:6 (describing Figure 2 as “depicting how methods of the invention allow one to detect, during analysis, whether the integrity of a patient’s sample was maintained during a multiplex sequencing reaction”).

The '852 patent states that its invention “is especially useful in multiplex next-generation sequencing applications in which errors can have a significant impact on results.” *Id.* at 2:12–14.

D. Illustrative Claim

Of the challenged claims, only claim 1 is independent. Claim 1 is illustrative of the claimed subject matter and is reproduced below.

1. A method for reducing template cross-over error introduced during sequencing workflow, the method comprising the steps of

incorporating at least two members of a plurality of identifier sequences into template nucleic acids obtained from at least two different samples, wherein said two members constitute a distinct pair associated with said template;

combining said nucleic acid templates into a single sample;

amplifying said nucleic acid templates on a surface of a flow cell thereby to form clusters, at least one of which comprises a chimeric sequence comprising a combination of identifier sequences that are different than any of said distinct pairs;

sequencing amplicons obtained in said amplifying step; and

discarding sequence reads obtained in said sequencing step that contain said combination of identifier sequences that are different than any of said distinct pairs, thereby to reduce cross-over error.

Ex. 1001, 20:65–21:19.

E. Instituted Challenges to Patentability

We instituted trial to determine whether the challenged claims are unpatentable based on the following bases:

Claims Challenged	35 U.S.C. § ²	References
1–8	103	Parameswaran, ³ Gloor ⁴
1–8	103	Parameswaran, Bentley ⁵

In support of their respective positions, Petitioner relies on the Declarations of Gregory B. Gloor, Ph.D. (Exs. 1002, 1046) and Paul T. Spellman, Ph.D. (Exs. 1003, 1054); and Patent Owner relies on the Declaration of Viktor Adalsteinsson, Ph.D. (Ex. 2001).

II. ANALYSIS

A. Principles of Law

To prevail in this *inter partes* review, Petitioner “shall have the burden of proving a proposition of unpatentability by a preponderance of the evidence.” 35 U.S.C. § 316(e); 37 C.F.R. § 42.1(d).

A claim is unpatentable for obviousness if “the differences between the subject matter sought to be patented and the prior art are such that the

² The Leahy-Smith America Invents Act (“AIA”), Pub. L. No. 112-29, 125 Stat. 284, 287–88 (2011), amended 35 U.S.C. § 103, effective March 16, 2013. On its face, the ’852 patent claims priority to a provisional application filed on December 23, 2010. Ex. 1001, code 60. For purposes of this proceeding, the parties apply this date as the effective priority date. *See* Pet. 17; PO Resp. 8. We do the same. Accordingly, we apply the pre-AIA version of § 103.

³ P. Parameswaran, et al., *A Pyrosequencing-Tailored Nucleotide Barcode Design Unveils Opportunities for Large-Scale Sample Multiplexing*, 35 NUCLEIC ACIDS RSCH, 19(e130) (2007) (Ex. 1004, “Parameswaran”).

⁴ G.B. Gloor, et al., *Microbiome Profiling by Illumina Sequencing of Combinatorial Sequence-Tagged PCR Products*, arXiv:1007.5075v1 (2010) (Ex. 1005, “Gloor”).

⁵ D.R. Bentley, et al., *Accurate Whole Human Genome Sequencing Using Reversible Terminator Chemistry*, 456 NATURE 6, 53–59 (2008) (Ex. 1006, “Bentley”).

subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains.” 35 U.S.C. § 103 (pre-AIA); *see also KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) when in evidence, objective evidence of nonobviousness. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

An invention is not obvious just because all of the elements that comprise the invention were known in the prior art. *KSR*, 550 U.S. at 418 (“[A] patent composed of several elements is not proved obvious merely by demonstrating that each of its elements was, independently, known in the prior art.”). Rather, a party that petitions the Board for a determination of unpatentability based on obviousness must show that “a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.” *In re Magnum Oil Tools Int’l, Ltd.*, 829 F.3d 1364, 1381 (Fed. Cir. 2016) (citations omitted).

In an obviousness determination, the presence or absence of a motivation to combine references, the presence or absence of a reasonable expectation of success, as well as what a reference teaches and whether it teaches toward or away from the claimed invention, are all questions of fact. *Par Pharm., Inc. v. TWI Pharm., Inc.*, 773 F.3d 1186, 1196–97 (Fed. Cir. 2014).

We analyze the instituted grounds of unpatentability in accordance with these principles.

B. Level of Ordinary Skill in the Art

In determining the level of ordinary skill in the art, various factors may be considered, including the “type of problems encountered in the art; prior art solutions to those problems; rapidity with which innovations are made; sophistication of the technology; and educational level of active workers in the field.” *In re GPAC, Inc.*, 57 F.3d 1573, 1579 (Fed. Cir. 1995). Furthermore, the prior art itself can reflect the appropriate level of ordinary skill in the art. *Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001).

Petitioner asserts that a person of skill in the art (“POSA”) “would have had (i) a Ph.D. in molecular biology, genetics, bioinformatics, or a related field and at least two years of experience in high-throughput sequencing technologies or (ii) a Master’s degree in one of the same fields with at least four years of experience in high-throughput sequencing technologies.” Pet. 17 (citing Ex. 1003 ¶¶ 19–20).

Patent Owner states that it adopts Petitioner’s proposal for a POSA for purposes of this proceeding. PO Resp. 13 (citing Ex. 2001 ¶¶ 19–22).

We adopt Petitioner’s proposed definition of the skill level as it is consistent with the prior art’s demonstration of the level of ordinary skill in the art at the time of the invention. We find both Dr. Spellman and Dr. Adalsteinsson are qualified to opine from the perspective of a POSA, because each has at least ordinary skill in the art. *See* Ex. 1003 ¶¶ 4–15; Ex. 2001 ¶¶ 4–9.

C. *Claim Construction*

In an *inter partes* review, we construe a claim term “using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. [§] 282(b).” 37 C.F.R. § 42.100(b) (2020). Under that standard, the words of a claim “are generally given their ordinary and customary meaning,” which is “the meaning that the term would have to a person of ordinary skill in the art in question at the time of the invention, i.e., as of the effective filing date of the patent application.” *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–13 (Fed. Cir. 2005) (en banc).

Claim 1 recites “template cross-over error” in the preamble and “cross-over error” in the body of the claim. Petitioner points out that “[t]he specification mentions the term ‘crossover’ four times, each describing the formation of a chimeric molecule made from template nucleic acids derived from different samples.” Pet. 22–23 (citing Ex. 1001, 5:41–6:6, 6:53–7:38). Petitioner proposes that we construe the term “template cross-over error” to mean “sequences of chimeric molecules made from template nucleic acids derived from different samples.” *Id.* at 23 (citing Ex. 1003 ¶ 122).

Patent Owner argues that Petitioner’s proposal “improperly broadens from the consistent descriptions in the ’852 Patent explaining that the invention addresses error caused by chimeric sequences *created during*

cluster amplification.”⁶ PO Resp. 13. According to Patent Owner, the ’852 patent specification repeatedly discusses cross-over errors “arising from the formation of chimeras during amplification after sample pooling.” *Id.* at 14 (citing Ex. 1001, Fig. 2, 5:1–5, 5:29–28, 5:41–43; Ex. 2001 ¶¶ 26–30, 41–49). Thus, Patent Owner proposes that “template cross-over error” and “cross-over error” should be construed to mean “error from sequences of chimeric molecules created during cluster amplification and made from template nucleic acids derived from different samples.”

Each party contends that its own proposed construction is supported, whereas the other side’s is contradicted, by the intrinsic evidence and general knowledge of a POSA. *See* PO Resp. 13–19; Reply 3–6; Sur-reply 1–7. Based on the complete record, we find Patent Owner’s argument more persuasive. We explain the reasoning for our construction and address Petitioner’s arguments below.

As an initial matter, we note that, during the oral argument, counsel for Petitioner argued that the term “[c]ross-over error was used in the prior art to refer just to a mixture of signals.” Tr. 57:6–7. To the extent that this argument suggests that “cross-over error” is a term of art with a generally

⁶ The parties also dispute whether to include the word “error” in the construction. *See* PO Resp. 13 (Patent Owner arguing Petitioner’s proposed construction improperly “redefines ‘error’ as ‘sequences’ as opposed to error from such sequences”); Reply 6 (Petitioner contending that “repeating the term ‘error’ adds nothing”); Sur-reply 1 (Patent Owner arguing “Petitioner’s proposed construction eliminates the idea of ‘error’ entirely”). In any event, Petitioner does not “object to the inclusion of the word error in the construction” because “it doesn’t affect either party’s arguments.” Tr. 12:4–6. Thus, we focus our claim-construction analysis on whether the term “cross-over error” should be limited to chimeric molecules “created during cluster amplification.”

known meaning, on this record, we are not persuaded. Indeed, in the Petition, Petitioner argues that the term “cross-over error” requires express construction, in contrast to “all other terms,” for which “[t]he plain and ordinary meaning is applied.” *See* Pet. 22. This distinction suggests that “cross-over error” is not a particular term of art with an established definition. In addition, Petitioner does not point to, and we do not find, any prior art of record that uses the term “cross-over error.” Thus, we must determine the meaning of the term “to the ordinary artisan after reading the entire patent.” *Phillips*, 415 F.3d at 1321.

Petitioner argues that claim 1 recites “reducing template cross-over error ***introduced during sequencing workflow***” and “does not require reducing cross-over error created at any specific point in that workflow, much less ‘during cluster amplification.’” Reply 3 (citing Ex. 1054 ¶¶ 10–12). In addition, Petitioner points out that “claim 1 recites only that at least one cluster on the flow cell ‘***comprises*** a chimeric sequence,’” and does not require that the chimeric sequences are “created” during cluster amplification. *Id.* at 4.

Petitioner is correct that claim 1 does not explicitly recite cross-over error created at any specific point in the sequencing workflow. But, as Petitioner recognizes, the recited cross-over error is error from “sequences of chimeric molecules” that are “made from template nucleic acids derived from ***different samples***.” Pet. 23 (emphasis added). As such, we agree with Patent Owner that “the claimed chimeric sequence could not be formed prior to the ‘combining’ step (i.e. when different samples are combined.)” *See* Sur--reply 2.

Claim 1 recites “amplifying said nucleic acid templates on a surface of a flow cell thereby to form clusters, at least one of which comprises a chimeric sequence.” In other words, claim 1 requires the chimeric sequence be present in clusters on the flow cell. Thus, we also agree with Patent Owner that “the *only* step during which chimeras could form between two nucleic acids from different samples is during the amplifying step.” *See* Sur-reply 2.

Pointing to the ’852 patent specification stating “[w]ithout being limited by any particular theory or mechanism of action,” Petitioner contends that “Patent Owner’s proposal contradicts the specification, which identifies cluster amplification as merely one possible theory for chimera formation.” Reply 4 (citing Ex. 1001, 6:25–29, 6:64–7:1). We are not persuaded by this argument.

The ’852 patent states that “[w]ithout being limited by any particular theory or mechanism of action, the data in FIG. 3 suggest that the nucleic acid from sample A and the nucleic acid from sample B landed within a cluster radius on the solid support.” Ex. 1001, 6:25–29; *see also id.* at 6:64–7:1. It continues to explain that, because of this close proximity of the samples, “[i]n certain cases, a heteroduplex between sample A’s nucleic acid template and sample B’s nucleic acid template was formed” during amplification. *Id.* at 6:29–38. In other words, the ’852 patent discloses that the formation of a heteroduplex between nucleic acids is one mechanism for forming a chimeric molecule during amplification.

Dr. Spellman, Petitioner’s expert, agrees. *See* Ex. 2002, 99:20–23 (Dr. Spellman testifying that “heteroduplex is one way of biochemically getting to a chimeric sequence. There are others”); 100:7–10 (Dr. Spellman

confirming that “a heteroduplex is one mechanism for creating a chimeric sequence but not necessarily the only mechanism”).

Thus, read in context, we agree with Patent Owner that the language Petitioner relies on (i.e., “[w]ithout being limited by any particular theory or mechanism of action”) refers to the method by which a chimeric molecule forms, not the stage of the sequencing flow during which it forms. *See* Sur-reply 5 (citing Ex. 2001 ¶ 29); *see also* PO Resp. 17 (“That specific example provides one theory or mechanism of how a chimeric molecule might be formed during amplification after sample pooling, but other mechanisms might also arise during amplification to create a chimeric molecule.”) (citing Ex. 1001, 6:25–29; Ex. 2001 ¶ 29).

Petitioner asserts that “Patent Owner’s construction contradicts the general knowledge in the art” because “[i]t was known that chimeric sequences made of templates from different samples could arise at many stages during sequencing, not just during cluster amplification on a flow cell.” Reply 5 (citing Ex. 1054 ¶¶ 20–25). We are not persuaded by this argument either.

First, experts for both parties agree that not all chimeric sequences result in cross-over errors. *See* Ex. 1055, 91:18–92:5 (Dr. Adalsteinsson testifying that chimeras other than those resulting in template cross-over can be formed); Ex. 2002, 84:25–85:3 (Dr. Spellman testifying that “the creation of a chimeric molecule can look a lot like or be exactly equivalent to a template crossing-over error. Distinguishing them can be extremely difficult”).

Second, Dr. Adalsteinsson testified that even though “other chimeras can be formed and sequenced, too, . . . the invention is directed at reducing

template cross-over error, and that's what's reflected in the claims.”

Ex. 1055, 92:1–5. Dr. Spellman appears to agree, testifying that “a very large set of biochemical processes can create chimeric molecules in sequencing workflows or more broadly,” but the claimed method is “well suited to detect and correct” errors “from crossover between nucleic acids from different samples.” Ex 2002, 84:11–85:3.

Third, as explained above, Petitioner has not sufficiently shown that “cross-over error” is a term of art with well understood meaning. Thus, we must construe the term in view of the specification, which, as “the single best guide to the meaning of a disputed term,” is usually dispositive. *See Phillips*, 415 F.3d at 1315. In contrast, extrinsic evidence showing what was known about chimeric sequences in general is less persuasive. *See id.*

Petitioner relies on the specification for “mention[ing] the term ‘crossover’ four times, each describing the formation of a chimeric molecule made from template nucleic acids derived from different samples.” Pet. 22–23 (citing Ex. 1001, 5:41–6:6, 6:53–7:38). Patent Owner argues that “Petitioner’s proposed construction improperly broadens the claim language by including only one of the two qualities defining every instance of crossover error in the ’852 Patent.” Sur-reply 4. We agree with Patent Owner that Petitioner ignores that each reference to “cross-over” in the Specification describes the chimeric molecule as forming during cluster amplification. *See id.*

Petitioner does not dispute that “in each of the specification quotes Petitioner relies on for its construction, and in every figure in the patent, the chimeric molecule also forms **during cluster amplification.**” Sur-reply 4. Petitioner, however, emphasizes that we should not “read limitations from

the embodiments in the specification into the claims.” Reply 4 (citing *Hill-Rom Servs., Inc. v. Stryker Corp.*, 755 F.3d 1367, 1371 (Fed. Cir. 2014)).

It is well recognized that “the distinction between using the specification to interpret the meaning of a claim and importing limitations from the specification into the claim can be a difficult one to apply in practice.” *Phillips*, 415 F.3d at 1323. “[T]he line between construing terms and importing limitations”, however, “can be discerned with reasonable certainty and predictability if [our] focus remains on understanding how a person of ordinary skill in the art would understand the claim terms.” *Id.* Ultimately, we construe claim terms to “capture the scope of the actual invention more accurately than either strictly limiting the scope of the claims to the embodiments disclosed in the specification or divorcing the claim language from the specification.” *Id.* at 1324. Here, we are tasked with resolving the dispute between the parties as to the scope of the source of the claimed cross-over error. *See* Sur-reply 2.

Although claim 1 does not explicitly recite cross-over error is created during cluster amplification, the specification shows otherwise. It repeatedly, consistently, and exclusively describes cross-over errors as arising from the formation of chimeric molecules during the amplification step of the sequencing workflow. *See* Ex. 1001, Fig. 1, Fig. 2, 4:29–52 (describing “improper molecular overlap via formation of a heteroduplex between patient A’s nucleic acid template and patient B’s nucleic acid template” “[w]hen the amplification process begins”), 5:29–58 (the same), 6:25–44 (the same), 6:64–7:23 (the same).

The Federal Circuit has explained that a patent’s repeated and consistent description of a claim term may inform its construction. *See Poly-Am., L.P. v. API Indus., Inc.*, 839 F.3d 1131, 1137 (Fed. Cir. 2016) (concluding that the district court’s claim construction “does not involve importing limitations from embodiments described in the specification” because “[e]very embodiment described in the specification has inwardly extended short seals and every section of the specification indicates the importance of inwardly extended short seals. These two facts provide together a proper reason to limit the claims in this way.”); *Profectus Tech. LLC v. Huawei Techs. Co.*, 823 F.3d 1375, 1381 (Fed. Cir. 2016) (concluding that “[w]hen read in view of the specification, the claims do not permit the expansive construction” because “every embodiment disclosed in the specification” includes the limiting feature).

The specification also expressly states that “the chimeric molecule is formed early in the amplification process.” Ex 1001, 4:57–58, 6:46–47. In contrast, the specification does not mention cross-over error created at any other point in a sequencing workflow. *See* Tr. 10:6–10 (Petitioner arguing that “it is possible” that cross-over errors could occur before the combining step, due to “just some kind of contamination between sample”). We agree with Patent Owner that Petitioner’s construction is too broad and not commensurate in scope with the specification because “Petitioner’s construction, an event occurring between two template molecules of different samples **at any time**, including before a multiplex sequencing reaction workflow even begins (e.g., cross-contamination between sample tubes during sample-gathering or pipetting steps), could cause formation of a chimeric sequence giving rise to cross-over error.” *See* PO Resp. 19 (citing

Ex. 2001 ¶¶ 24–26) (emphasis added). We agree with Patent Owner that such a broad construction would “divorc[e] the claim language from the specification.” *See Phillips*, 415 F.3d at 1324. Instead, a construction of “cross-over error” that limits its creation during the amplification step “is required to tether the claims to what the specification[] indicate[s] the inventor actually invented.” *Retractable Techs., Inc. v. Becton, Dickinson & Co.*, 653 F.3d 1296, 1305 (Fed. Cir. 2011).

In sum, in view of the specification, we agree with Patent Owner that “cross-over error” means “error from sequences of chimeric molecules created during cluster amplification and made from template nucleic acids derived from different samples.” *See* PO Resp. 13.

Claim terms need only be construed to the extent necessary to resolve the controversy. *Wellman, Inc. v. Eastman Chem. Co.*, 642 F.3d 1355, 1361 (Fed. Cir. 2011). On this record and for purposes of this Decision, we see no need to expressly construe any other claim term to resolve the issues before us.

D. Alleged Obviousness over Parameswaran and Gloor

Petitioner asserts that claims 1–8 would have been obvious over the combination of Parameswaran and Gloor. Pet. 23–51. After reviewing the entire record developed at trial, and as explained below, we determine Petitioner has not shown by a preponderance of the evidence that the challenged claims would have been obvious over the asserted prior art.

1. Parameswaran

Parameswaran is titled “A pyrosequencing-tailored nucleotide barcode design unveils opportunities for large-scale sample multiplexing.” Ex. 1004, 1. Parameswaran uses 454/Roche Genome Sequencers for

sequencing. *Id.* The technology of these pyrosequencers is based on the detection of pyrophosphates released during DNA synthesis. *Id.*

Parameswaran describes “a high-information-content barcoding approach in which each sample is associated with two uniquely designed, 10-nucleotide barcodes.” *Id.* at 2. According to Parameswaran, “[t]he presence of these assigned barcodes allow[s] for independent samples to be pooled together for sequencing, with subsequent bioinformatic segregation of the pyrosequencer output.” *Id.* In addition to barcodes, Parameswaran incorporated forward and reverse cloning linkers into nucleic acids. *Id.* at 2–3.

Figure 1 of Parameswaran, reproduced below, depicts the design of forward and reverse primers. *Id.* at 3.

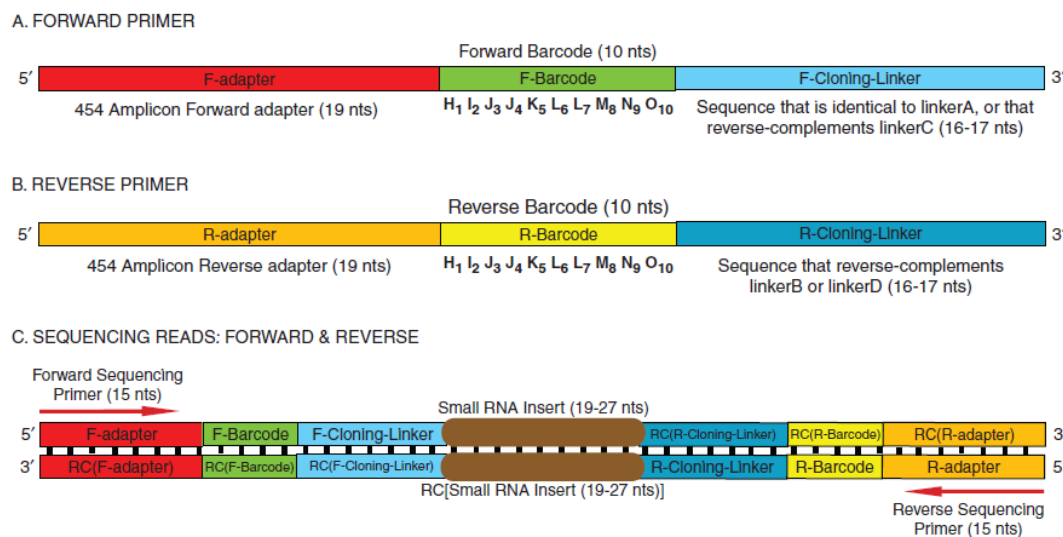


Figure 1. Design of forward and reverse primers. The synthesized primers are 45–46 nt long. (A and B) The template for the primers is: 454-Adapter::Barcode::Linker-Primer. Individual specifications for forward and reverse barcodes are also indicated. (C) Diagrammatic representation of the GS20 forward and reverse sequencing reads. ‘RC’ stands for Reverse Complement. Teeth denote base pairing.

Id. at 3. Parameswaran’s Figure 1, above, shows that a forward primer comprises the Roche-specific forward adapter (“F-adapter”), the forward barcode (“F-Barcode”), and the forward cloning linker (“F-Cloning-Linker”). *Id.* It also shows that a reverse primer comprises the

Roche-specific reverse adapter (“R-adapter”), the reverse barcode (“R-Barcode”), and the reverse cloning linker (“R-Cloning-Linker”). *Id.*

Parameswaran describes expanding the 10-nucleotide barcodes to “barcode motifs,” which included three additional nucleotides from linkers adjoining the barcodes, to maximize the uniqueness of dual-indexing and facilitate definitive sample assignments. *Id.* at 7. Parameswaran’s study aimed to determine “how much of the (fixed) linker sequence will be required in addition to the (variable) barcode for definitive sample assignment.” *Id.* at 7–8. Ultimately, “a search motif of 13 nt was determined to be ideal for our studies in minimizing false-discovery rate.” *Id.*

Parameswaran reports that using 13-nt barcode motifs, approximately 94% of sequences were identified to have “perfect forward and reverse barcode motifs.” *Id.* at 7. Parameswaran subdivided the sequences having a perfect match to only one of the expected barcode motifs into three classes. *Id.* One of the three classes includes those having “perfect identity to an unexpected motif, resulting in an unassigned 5’–3’ motif combination (perfect, unexpected occurrence).” *Id.* Parameswaran reports that “[t]he false-discovery rate, defined by the percentage of ‘perfect and unexpected’ occurrences, was observed to decrease to <0.005%.” *Id.* at 8.

Parameswaran speculates that its pyrosequencer-tailored barcodes may also be compatible with other sequencing platforms. *Id.* at 2, 8. Parameswaran specifies that Illumina’s platform is one of three major high-throughput sequencing platforms. *Id.* at 1.

2. Gloor

Gloor is titled “Microbiome profiling by Illumina sequencing of combinatorial sequence-tagged PCR products.” Ex. 1005, 1. Gloor describes

“a low-cost, high-throughput microbiome profiling method that uses combinatorial sequence tags attached to PCR primers that amplify the rRNA V6 region.” *Id.*; *see also id* at 2 (reporting “the methods used to perform microbiome analysis of the V6 region of 272 clinical samples using the Illumina sequencing technology”). Gloor describes sequencing amplified PCR products “using an Illumina paired-end protocol to generate millions of overlapping reads.” *Id.* at 1. According to Gloor, “[c]ombinatorial sequence tagging can be used to examine hundreds of samples with far fewer primers than is required when sequence tags are incorporated at only a single end.” *Id.*

Gloor added primers and sequence tags to nucleic acids prior to sequencing. *Id.* at 3–6. Gloor “var[ied] the length of the tags between 3 and 6 nucleotides long” and “chose sequence tags to ensure all 4 nucleotides were represented in each of the first four positions of the primers using parameters similar to those in the barcrawl program.” *Id.* at 5. Gloor describes developing custom programs “to identify the overlapping segments of the forward and reverse reads” and to extract “overlapping reads with proper primer sequences and correct sequence tags.” *Id.* at 7. Gloor inspected sequences with “incorrect sequence tags” to determine the sources of the errors. *Id.* at 7–8.

3. Analysis

We focus our analysis on independent claim 1. Petitioner argues that the combination of Parameswaran and Gloor teaches or suggests each limitation of claim 1. Pet. 23–36. Petitioner also contends that an ordinarily skilled artisan would have been motivated combine the teachings of the two

references and would have had a reasonable expectation of success in doing so. *Id.* at 36–45.

Patent Owner disputes Petitioner’s contentions. PO Resp. 20–51. Patent Owner argues that Petitioner fails to show a motivation to combine Parameswaran and Gloor⁷ to arrive at the claimed invention with a reasonable expectation of success. *Id.* at 20–41. Patent Owner also asserts that the combination does not teach or suggest the discarding step of the claimed method. *Id.* at 42–44. We find Patent Owner’s arguments more persuasive.

a. The Step of “Discarding . . . Thereby to Reduce Cross-Over Error”

Claim 1 recites “discarding sequence reads obtained in said sequencing step that contain said combination of identifier sequences that are different than any of said distinct pairs, thereby to reduce cross-over error.” Petitioner argues that Parameswaran discloses this step.⁸ Pet. 34–35.

In Parameswaran, on Roche454 platform, using 13-nt barcode motifs, about 94% of the sequences had perfect forward and reverse barcode motifs,

⁷ Patent Owner challenges Gloor’s prior art status. PO Resp. 45–51. As explained below, we agree with Patent Owner that Petitioner has not met its burden to show the combination of Parameswaran and Gloor renders the challenged claims obvious. Thus, we do not need to decide whether Petitioner has met its burden to show Gloor is prior art.

⁸ Petitioner acknowledges that “because *Gloor*’s pairs of identifiers were prepared in a combinatorial manner (i.e., non-unique dual-indexing), the pair of identifiers found in these chimeric sequences may nevertheless match one of the pairs used for other nucleic acid templates.” Pet. 33 (citing Ex. 1003 ¶ 149); *see also id.* at 36, 38 (the same). Thus, we agree with Patent Owner that “Gloor’s use of combinatorial indexes meant that its method could not have unearthed the cross-over error that Petitioner alleges a POSA would seek to reduce.” PO Resp. 34 (citing Ex. 2001 ¶¶ 74–84, 124).

while about 5–6% had perfect barcode motifs only at one end. Ex. 1004, 7. A subset of this 5–6% of the sequences contained misassignment, i.e., “unexpected perfect pairings of unmatched forward and reverse barcodes.” *Id.* at 1.

Petitioner points to Parameswaran’s teaching of mismatched barcode motifs, where “the forward identifier was incorrectly paired with the wrong reverse identifier.” Pet. 32 (citing Ex. 1004, 8, Abstract; Ex. 1003 ¶ 147). According to Petitioner, Parameswaran’s “sequences with mismatched barcode motifs include chimeric sequences resulting from cross-over events.” *Id.* (citing Ex. 1003 ¶ 148). Specifically, Petitioner argues that the “unexpected perfect pairings of unmatched forward and reverse barcodes” are chimeric sequences resulting from cross-over error. *Id.* at 25–26 (citing Ex. 1003 ¶¶ 128–131; Ex. 1004, 7–8).

Petitioner contends that Parameswaran “distinguished sequences obtained in the sequencing step that contained combinations of identifier sequences that were different than any of assigned combinations of identifier sequences.” Pet. 34 (citing Ex. 1003 ¶ 153). According to Petitioner, an ordinarily skilled artisan “would have understood that Parameswaran disclosed a method in which these chimeric sequences could be discarded, and thus, reduce cross-over error.” *Id.* at 35 (citing Ex. 1003 ¶ 154). Pointing to Parameswaran’s language of “false-discov[ies]” and “misassignment,” Petitioner argues that “[a] POSA would have understood that these false discoveries could be excluded to increase accuracy.” *Id.* (citing Ex. 1003 ¶ 154; Ex. 1004, 7–8). Petitioner asserts that “*Parameswaran* states that, by lowering the false-discovery rate, ‘any sequence with both 5’ and 3’ barcode motifs will be placed with reasonable accuracy into the correct bin.’” *Id.*

(citing Ex. 1003 ¶ 154; Ex. 1004, 8). Thus, Petitioner concludes that Parameswaran teaches a method for reducing template cross-over error. *Id.* at 26. We are not persuaded by Petitioner’s argument.

Parameswaran states that “[p]erfect, ‘unexpected’ occurrences may be attributed to motifs with rare coincidental sequence matches (due to multiple fortuitous errors during barcode synthesis or sequencing), to contamination in PCR primer stocks, or to cross-contamination of template DNAs in the second round of PCR.” Ex. 1004, 7.

We agree with Patent Owner that “the coincidental sequence matches are a product of barcode design itself, not any potential cross-over error during the sequencing workflow.” PO Resp. 33 (citing Ex. 2001 ¶¶ 66, 101, 120, 121). We also agree with Patent Owner that “contamination in PCR primer stock cannot be cross-over error because PCR primers are added prior to any pooling, amplification, or sequencing step in a nucleic acid workflow.” *Id.* (citing Ex. 2001 ¶¶ 122–123). We further agree with Patent Owner that cross-contamination in the second round of PCR is not cross-over error because it “occurs ***before any samples are pooled together***, amplified, or sequenced.” *Id.* at 34 (citing Ex. 2001 ¶¶ 122–123); *see also* Ex. 1004, 3–4 (describing two rounds of amplification of each individual sample before pooling twenty-five samples together).

Petitioner does not dispute these assertions. Indeed, Dr. Spellman testified that the “likely interpretation” for Parameswaran’s “large numbers of perfect unexpected barcode motifs” is that they “were the result of sequencing errors, not barcode swapping.” Ex. 2002, 168:6–17. Dr. Spellman also agreed that “that’s why adding nucleotides from the cloning linkers essentially solved the problem.” *Id.*

Thus, on this record, we are persuaded by Patent Owner’s argument that Parameswaran’s “unexpected perfect pairings of unmatched forward and reverse barcodes” are not cross-over error. PO Resp. 31 (citing Ex. 2001 ¶¶ 116–123).

In addition, as Petitioner acknowledges, Parameswaran merely “distinguished,” but did not “discard” sequences with mismatches. Pet. 34. Parameswaran states that “[t]he low false-discovery rate for perfect matches strengthens the argument that any sequence with both 5’ and 3’ barcode motifs will be placed with reasonable accuracy into the correct bin.” Ex. 1004, 8. But, because Parameswaran achieved a “remarkably low misassignment rate,” it did not discard, but included any sequence with a matched barcode. *Id.* Indeed, Parameswaran explains that “[f]or our purposes, all sequences with a match to one of the two expected sequence motifs were considered as true members of the corresponding dataset.” *Id.*

In its Reply, Petitioner emphasizes Parameswaran’s language “[f]or our purposes.” Reply 14 (citing Ex. 1004, 8). According to Petitioner, “[a] POA would have understood from Parameswaran that chimeric sequences could be kept or discarded, depending on the sequencing ‘purpose.’” *Id.* (citing Ex. 1054 ¶¶ 79–82).

Petitioner is asking us to deem unpatentable a claim based on its reading of “for our purposes,” a broad, undefined statement, to mean something specific—a suggestion to discard perfect unexpected pairings, which Parameswaran explicitly did not do. *See* Ex. 1004, 8. Petitioner has not presented sufficient evidence and persuasive argument for us to do so.

Petitioner also contends that Parameswaran refers to sequences with “*expected* sequence motifs,” and not “perfect, *unexpected* occurrences.”

Reply 14 (citing Ex. 1004, 7). We disagree. Parameswaran included “all sequences with a match to *one of the two expected* sequence motifs.” Ex. 1004, 8 (emphasis added). In other words, sequences with one expected sequence motif and another unexpected sequence motif were included. Indeed, Parameswaran explicitly states that only “sequences [that] had no perfect matches to any of the sequence motifs (forward or reverse) were discarded.” Ex. 1004, 8.

Pointing to Parameswaran’s supplementary table, Petitioner also argues that sequences with mismatched barcode motifs were greyed out, “signifying that they were discarded.” *Id.* at 15 (citing Ex. 1004, 23, 29; Ex. 1054 ¶¶ 84, 85, 89). According to Petitioner, “[t]hese tables indicate that *Parameswaran* ‘discarded’ chimeric sequences to reduce cross-over error,” or at least “reinforce that it would have been obvious to do so.” *Id.* at 15 (citing Ex. 1054 ¶ 89). We are not persuaded by this argument either.

Dr. Spellman testified that in the supplementary table, the discarded sequences correspond to the four blank cells in the bottom right of the table, not to the greyed-out cells. Ex. 2015, 102:3–104:19. As Patent Owner points out, “[n]either Dr. Spellman nor Petitioner explain[s] why the grayed boxes they highlight were not simply left blank if those sequences were actually discarded, like the four boxes in the data table corresponding to sequences *Parameswaran* actually states were discarded.” Sur-reply 22.

In sum, despite labelling sequences with “mismatched 5’ and 3’ motifs (i.e. sequences that do not have corresponding 5’ and 3’ motifs)” as “false-discover[ies]” with “misassignment[s],” *Parameswaran* included, and did not discard, the sequences with perfect, but unexpected, barcode motifs. Ex. 1004, 8 (stating “all sequences with a match to one of the two expected

sequence motifs were considered as true members of the corresponding dataset). Thus, Parameswaran’s method does not teach or suggest “discarding sequence reads obtained in said sequencing step that contain said combination of identifier sequences that are different than any of said distinct pairs, thereby to reduce cross-over error.” See PO Resp. 41 (citing Ex. 2001 ¶¶ 116–123, 135–137).

*b. Reasonable Expectation of Success*⁹

Petitioner argues that

a POSA would have had a reasonable expectation of success in combining Parameswaran and Gloor at least because a POSA would have recognized that (i) performing the unique dual-indexing method taught by *Parameswaran* using Illumina sequencers and flow cells would simply require swapping the Roche 454 adapters for Illumina-specific adapters; (ii) Illumina sequencing enabled read lengths appropriate to sequence the template nucleic acids disclosed in *Parameswaran*; and (iii) *Parameswaran*’s unique dual-barcoding approach would desirably reduce cross-over error on an Illumina sequencer.

Pet. 41–42 (citing Ex. 1003 ¶¶ 165–176).

Patent Owner contends that even if Petitioner’s arguments (i) and (ii) are true, “neither of those assertions relates to cross-over error or a POSA’s expectation of reducing such error as claimed.” PO Resp. 40. We agree with Patent Owner.

“The reasonable expectation of success requirement refers to the likelihood of success in combining references to meet the limitations of the claimed invention.” *Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*,

⁹ Because, as explained below, we find Petitioner has not met its burden to show reasonable expectation of success, we do not need to address whether Petitioner has shown sufficient reason to combine the asserted references.

821 F.3d 1359, 1367 (Fed. Cir. 2016); *see also Allergan, Inc. v. Apotex Inc.*, 754 F.3d 952, 966 (Fed. Cir. 2014) (stating that “failure to consider the appropriate scope of the . . . patent’s claimed invention in evaluating the reasonable expectation of success . . . constitutes a legal error”). Here, the claimed invention is a method for reducing template cross-over error. Thus, we must determine whether, in December 2010, an ordinarily skilled artisan would have reasonably expected that the combination of Parameswaran and Gloor would reduce template cross-over error.

Petitioner points out that Parameswaran states that its barcoding approach is compatible “with other sequencing platforms” and specifically identifies Illumina sequencers as one of the “major high-throughput sequencing platforms.” Pet. 44–45 (citing Ex. 1004, 1, 2, 8). Thus, Petitioner argues that “a POSA would have expected the unique dual-barcode approach described in *Parameswaran* to successfully reduce cross-over error on an Illumina sequencer.” *Id.* at 44 (citing Ex. 1003 ¶ 172).

Patent Owner asserts that “[a]t the time of the ’852 Patent, a POSA would not have known that crossover error was problematic for NGS, particularly for sequencers that amplify on a flow cell like Illumina’s.” PO Resp. 29 (citing Ex. 2001 ¶¶ 105–115); *see also id.* at 40 (“None of Parameswaran, Gloor, or the prevailing views in the art at the time of the invention, identified **template cross-over error** as a source of error that Parameswaran’s indexes could address or reduce at the time of the patented invention.”). According to Patent Owner, “a POSA, having no knowledge that cross-over error was even a problem for NGS, and particularly for Illumina sequencers like Gloor’s, would have no expectation to ‘reduce cross-over error’ as claimed by incorporating Parameswaran’s UDIs into

Gloor’s Illumina sequencer approach.” *Id.* at 39. We find Patent Owner’s arguments more persuasive.

As explained above, we find that Parameswaran’s “unexpected perfect pairings of unmatched forward and reverse barcodes” are not cross-over errors, and that Parameswaran does not teach or suggest the discarding step to reduce cross-over error. *See supra* Section II.D.3.a. Below, we discuss the parties’ arguments on whether cross-over errors resulting from chimeric molecules were known generally.

Petitioner cites several references to support its argument that cross-over errors resulting from chimeric molecules were known. Pet. 6–7 (citing Ex. 1017, 1; Ex. 1018, 1; Ex. 1019; 1; Ex. 1020, 7; Ex. 1033, 8–9; Ex. 1034, 10, 13). As explained above, we construe “cross-over error” to mean “error from sequences of chimeric molecules created during cluster amplification and made from template nucleic acids *derived from different samples*.” *See supra* Section II.C (emphasis added).

Patent Owner contends that some of the references Petitioner cites “discuss only potential chimeras formed *within samples* (e.g., during standard PCR), do not utilize any tagged nucleic acids, and/or generally provide only potential mechanisms that could create chimeras in standard PCR,” whereas those that “actually address NGS do not discuss the claimed cross-over error or chimeric sequences giving rise to such error.” *See* PO Resp. 30 (citing Ex. 1017, 5; Ex. 1018, 3–4; Ex. 1019, 1; Ex. 1020, 1; Ex. 1033, 9; Ex. 1034, 10; Ex. 2001 ¶¶ 105–115). We agree with Patent Owner.

For example, Dr. Adalsteinsson testifies that

Lahr¹⁰ describes the “percentages of chimeras formed per reaction” when PCR is performed under varied conditions and concludes that “[c]apturing the full diversity *within a sample* requires a combination of multiple PCR reactions that have been performed under chimera-reducing conditions.” EX1017 (Lahr), 5. Template nucleic acids in Lahr are not tagged, and the “worrisome aspect of PCR-based studies” to which Lahr refers is the chimera formation affecting the determination of the “actual genetic diversity *of a sample*.” EX1017 (Lahr), 1. A POSA would therefore understand Lahr to reference chimeras within a given sample, not between two different samples

Ex. 2001 ¶ 107 (footnote added). We find Dr. Adalsteinsson’s interpretation of Lahr reasonable. We similarly credit Dr. Adalsteinsson’s testimony regarding Exhibits 1018, 1019, and 1020 that those references only teach chimeras formed within samples in standard PCR, and not derived from different samples, as required even under Petitioner’s proposed construction of “cross-over error.” *See id.* ¶¶ 108–110.

Petitioner argues that “chimeric molecules were particularly problematic for multiplex sequencing, causing misassignment of sequences to incorrect samples and compromising sequencing accuracy.” Pet. 7. Citing Creer,¹¹ Petitioner asserts that NGS was “the ideal ‘breeding ground’” for chimeric sequences. *Id.* (quoting Ex. 1034, 10). Petitioner, however, has not

¹⁰ Lahr et al., Reducing the Impact of PCR-Mediated Recombination in Molecular Evolution and Environmental Studies Using a New-Generation High-Fidelity DNA Polymerase, 47 BIOTECHNIQUES 4, 857–66 (2009) (Ex. 1017, “Lahr”).

¹¹ Creer et al., *Ultrasequencing of the Meiofaunal Biosphere: Practice, Pitfalls and Promises*, 19 MOLECULAR ECOLOGY (Suppl. 1) 4–20 (2010) (Ex. 1034, “Creer”).

sufficiently shown that the chimeras discussed in Creer are the same as the claimed chimeric sequences.

Creer teaches ultrasequencing of the meiofaunal biosphere using Roche 454 sequencing platforms. Ex. 1034, 1. It “report[s] two examples of environmental metagenetic nuclear small subunit 18S (nSSU) analyses of marine and tropical rainforest habitats and provide[s] critical appraisals of the level of putative recombinant DNA molecules (chimeras) in metagenetic data sets.” *Id.*

Creer explains that *in vitro* recombination of homologous DNA leads to “disconcerting” proportion of putative chimeras. Ex. 1034, 10. According to Creer, “[t]he actual species richness and taxonomic composition of the samples may contribute to the level of chimera formation.” *Id.* Indeed, Creer states that “[i]n many ways, metagenetic ultrasequencing experiments are the ideal ‘breeding ground’ for recombinant DNA molecules” because “[t]hey are based on the amplification of homologous regions.” *Id.* In contrast, the claimed chimeric sequence comprises “a combination of identifier sequences that are different than any of said distinct pairs.” Thus, we find that chimeras generated in Creer’s metagenetic ultrasequencing experiments are different from the claimed chimeric sequences formed during NGS.¹² See Ex. 2001 ¶ 112 (“Creer does not propose that any chimeric sequences have formed between template nucleic acids *of different samples*. Rather, the suggestions they provide refer to the individual ‘environmental PCRs’ *of each sample*.”).

¹² This finding is further supported by the fact that, when discussing the chimeras, Creer cites papers published in 1997 and 2001, which predates the emergence of NGS systems.

We similarly find persuasive Dr. Adalsteinsson's testimony that Dai¹³ does not show whether the "false chimeric sequences" discussed therein are cross-over errors from different samples. *See* Ex. 2001 ¶ 113. Instead, Dai suggests that "it is likely that many such 'novel' genomic structural changes are caused by problems in library preparation." Ex. 1033, 9 ("[M]ost of the genomic structural changes revealed by paired-end sequencing for this subject must be the result of poor sequencing library preparation."). Thus, we agree with Patent Owner that, although Dai and Creer examine chimeric sequences in NGS data, neither describes cross-over errors. *See* PO Resp. 30; Ex. 2001 ¶ 111.

In its Reply, Petitioner relies on other references, including Campbell.¹⁴ Reply 7 (citing Ex. 1057). Petitioner argues that Campbell, using an Illumina sequencer, "detected and excluded nearly 1.2 million erroneous chimeric sequences—including PCR-induced artifacts formed 'within the PCR colony' on Illumina's flow cells." *Id.* (citing Ex. 1054 ¶¶ 40–44; Ex. 1057, 1, 2, 7). According to Petitioner, Campbell is one of "numerous prior-art references disclosing the problem of chimeric sequences in Illumina sequencing." *Id.* We are not persuaded by this argument.

Campbell "used massively parallel sequencing to generate sequence reads from both ends of short DNA fragments derived from the genomes of

¹³ Dai, et al., *NGSQC: Cross-Platform Quality Analysis Pipeline for Deep Sequencing Data*, 11 BMC GENOMICS 11(Suppl 4):S7 (2010) (Ex. 1033, "Dai").

¹⁴ Campbell et al., *Identification of Somatically Acquired Rearrangements in Cancer Using Genome-Wide Massively Parallel Paired-End Sequencing*, 40 NATURE GENETICS 722–29 (2008) (Ex. 1057, "Campbell").

two individuals with lung cancer.” Ex. 1057, 1. Campbell states that “[t]here were ~1.2 million reads from [cell line] NCI-H2171 and 770,000 from NCI-H1770 in which mapping of the two ends to the genome suggested a possible rearrangement.” Ex. 1057, 2. Campbell summarized the confirmed rearrangements in Table 1. *Id.* at 3. As Patent Owner points out, however, “Campbell never once mentions a chimera or a chimeric sequence.” Sur-reply 9. Instead, Campbell lists deletions, inversions, inverted duplications and tandem duplications as examples of the confirmed rearrangements. Ex. 1057, 3. From Dr. Spellman’s deposition testimony, we understand that these rearrangements are different from the claimed chimeric sequences. *See* Ex. 2015, 50:6–56:4. Neither Petitioner nor Dr. Spellman sufficiently explains which other rearrangements in Campbell support the statement that “about ‘1.2 million reads’ ‘suggested a possible rearrangement’—i.e., chimeric sequences.” Ex. 1054 ¶ 42 (citing Ex. 1057, 2). Thus, we are not persuaded that Campbell teaches the problem of chimeric sequences in Illumina sequencing.

Petitioner argues that other studies, such as Varela,¹⁵ “similarly detected and excluded erroneous chimeric sequences during Illumina sequencing, including PCR-induced artifacts formed ‘within the bridge amplified cluster.’” Reply 7 (citing Ex. 1066, 10). On this record, we are not persuaded the artifacts in Varela are the same as the claimed chimeric sequence.

¹⁵ Varela et al., Somatic Structural Rearrangements in Genetically Engineered Mouse Mammary Tumors, 11 GENOME BIOLOGY R100, 1–12 (2010) (Ex. 1066, “Varela”).

Varela teaches somatic structural rearrangements in genetically engineered mouse mammary tumors. Ex. 1066, 1. Varela does not mention chimeras. *See generally* Ex. 1066. Instead, Varela attributes its artifacts to “resulting from either intramolecular rearrangements generated during library amplification or mispriming of the sequencing oligonucleotide within the bridge amplified cluster.” *Id.* at 10. We are persuaded by Patent Owner’s argument that “intramolecular rearrangements” “cannot be a chimera of two **different** molecules,” and “mispriming of the sequencing oligonucleotide” “occurs **after** amplification on the flow cell.” Sur-reply 11–12 (citing Ex. 1066, 10).

Petitioner also alleges that Maher¹⁶ teaches “detecting candidate chimeras formed among templates ‘in close proximity on the flow cell.’” Reply 8 (citing Ex. 1061, 8, 16). Maher, however, teaches “a sensitive, high-throughput methodology to comprehensively catalog functional gene fusions in cancer.” Ex. 1061, 1. Maher explains that gene fusion is “[o]ne of the most common classes of genetic alterations.” *Id.* Thus, we agree with Patent Owner that “Maher detects **biological** chimeras (i.e., those naturally present in a sample when collected) as potential cancer markers” (Sur-reply 12), and not the chimeric sequence “comprising a combination of identifier sequences that are different than any of said distinct pairs,” as claimed.

In addition, Maher’s artifacts “in close proximity on the flow cells” have “identical mate pair reads.” Ex. 1061, 8; *see also* Ex. 2015,

¹⁶ Maher et al., *Chimeric Transcript Discovery by Paired-End Transcriptome Sequencing*, 106 PNAS (30) 12353–58 (2009) (Ex. 1061, “Maher”).

122:16–123:6 (Dr. Spellman confirming this teaching). Neither Petitioner nor Dr. Spellman sufficiently explain how an ordinarily skilled artisan would understand the artifacts with “identical mate pair reads” as the claimed chimeric sequences. *See* Ex. 2015, 125:17–20 (Dr. Spellman testifying that Maher does not describe “what an identical mate-pair read really is”). Instead, Maher states that “[t]hese duplicates were likely an artifact of the analysis pipeline.” Ex. 1061, 8; *see also* Ex. 2015, 126:3–11 (Dr. Spellman testifying that it is a “fair” interpretation that the duplicates in Maher “aren’t a situation where the original seeding molecule changes into two different molecules;” instead, “[i]t’s a situation where one original seeding molecule gets duplicated to be read as two different copies of the same starting sequence”). Thus, we are not persuaded that Maher teaches detecting the claimed chimeric sequences. *See* Reply 8.

Petitioner further contends that Whiteford¹⁷ teaches “‘mixed clusters can be grown starting from more than one template’ in Illumina sequencing.” Reply 8 (citing Ex. 1067, 3). This teaching, however, does not support Petitioner’s argument that, at the priority date, “POSAs understood that many chimeric sequences arose on Illumina flow cells.” *See id.* This is because, as Dr. Spellman explained, a “mixed cluster” refers to the situation “in which the clusters completely overlap but maintain their identity.” Ex. 2015, 26:8–11. In a “mixed cluster,” Dr. Spellman continued, there would be “at every cycle the potential for two different bases or more being detected at each round of the chemistry,” “it’s hard for the machine to know

¹⁷ Whiteford et al., *Swift: Primary Data Analysis for the Illumina Solexa Sequencing Platform*, 25 BIOINFORMATICS (17) 2194–99 (2009) (Ex. 1067, “Whiteford”).

exactly what base is added to each cluster.” *Id.* at 26:11–27:3. Neither Petitioner nor Dr. Spellman sufficiently explains how an ordinarily skilled artisan would have understood such “mixed clusters” in Whiteford to be the same as the claimed chimeric sequences.

Petitioner argues that “[w]hen multiple samples were sequenced using NGS platforms, chimeric sequences were particularly problematic because they created ‘false’ sequences and decreased accuracy.” Reply 8. As support, Petitioner cites Gorzer¹⁸ and Smyth.¹⁹ *Id.* at 8–9.

Petitioner relies on Gorzer for teaching that “[w]hen PCR amplification is carried out using a mixture of two or more non-identical but similar genotype target sequences as templates, this can lead potentially to the formation of artificial chimeras between the amplification products.” *Id.* (quoting Ex. 1060, 1). The amplification in Gorzer refers to the “initial PCR step,” which “is required to amplify each genotype.” Ex. 1060, 1. Patent Owner argues that Gorzer’s chimeras formed during “initial PCR,” which was “prior to pooling or sequencing,” and thus, is not the claimed cross-over error. Sur-reply 13. We find Patent Owner’s argument more persuasive.

Indeed, to support its statement about the formation of artificial chimeras between the amplification products,” Gorzer cites two references published in 1990 and 1998, as well as “Lahr and Katz, 2009.” Ex. 1060, 1. The references published in 1990 and 1998 predates the NGS platforms, and

¹⁸ Gorzer et al., The Impact of PCR-Generated Recombination on Diversity Estimation of Mixed Viral Populations by Deep Sequencing, 169 J. VIROLOGICAL METHODS 248–52 (2010) (Ex. 1060, “Gorzer”).

¹⁹ Smyth et al., Reducing Chimera Formation During PCR Amplification to Ensure Accurate Genotyping, 469 GENE 45–51 (2010) (Ex. 1063, “Smyth”).

thus, cannot support Petitioner's reliance on Gorzer for allegedly showing chimeric sequences on NGS platforms.

The "Lahr and Katz" reference cited in Gorzer is the same as the Lahr reference Petitioner cites to support its argument that cross-over errors resulted from chimeric molecules were known. As explained above, we credit Dr. Adalsteinsson's testimony that Lahr discusses chimeras formed "within a given sample, not between two different samples." *See* Ex. 2001 ¶ 107. Thus, we are persuaded by Patent Owner's argument that Gorzer's chimeras were formed in standard PCR, and not during a flow cell sequencing workflow, as claimed. *See* Sur-reply 13.

Petitioner also cites Smyth to show that chimeric sequences were formed on NGS platforms. *See* Reply 8. Similar to Gorzer, when discussing chimeras, Smyth refers to references published before the commercial NGS platforms. *See* Ex. 1063, 1 (citing papers published in 1990, 1995, 1998, and 2002). Thus, we are persuaded by Patent Owner's argument that Smyth's chimeras were formed in standard PCR, and not during a flow cell sequencing workflow, as claimed. *See* Sur-reply 13.

We acknowledge that at the priority date, "many NGS platforms used PCR during sequencing, including bridge amplification on a flow cell." Reply 9 (citing Ex. 1003 ¶¶ 59–61; Ex. 1054 ¶¶ 49–51). Nonetheless, we are not persuaded that the claimed cross-over error on a flow cell was known because chimera formed in standard PCR was known. *See* Reply 9–10 (citing Ex. 1015, 9; Exs. 1017–1020). Instead, we find evidence proffered by Patent Owner tends to support its argument that cross-over error on Illumina sequencers was not appreciated until after the '852 patent's priority date. *See*

PO Resp. 35 (citing Exs. 2003, 2004, 2006, 2007); Sur-reply 14 (citing Exs. 2003, 2004).

Kircher²⁰ is a research article published online in October 2011. Ex. 2004, 1. Although Kircher was published ten months after the priority date of the '852 patent, it nonetheless can be used as evidence of the state of art existing at the time of the priority date. *See, e.g., Plant Genetic Sys., N.V. v. DeKalb Genetics Corp.*, 315 F.3d 1335, 1344 (Fed. Cir. 2003).

Using Illumina's sequencer, Kircher "developed a method that reveals the rate of sample misidentification within current multiplex sequencing experiments" by "introducing indexes into both library adapters (double indexing)." Ex. 2004, 1. Kircher acknowledges that "PCR can produce chimeras by recombining different templates molecules (a process often referred to as 'jumping PCR')." Ex. 2004, 2 (citing Exs. 2017–2020).

Indeed, as Petitioner points out, "before December 2010, the authors of EX2004 expressly warned of 'jumping PCR' between samples during Illumina sequencing." Reply 10 (citing Ex. 1015, 9). The reference Petitioner refers to, Exhibit 1015,²¹ cites Exhibit 1018 as support of occurrence of jumping PCR and suggests amplifying libraries "with a minimum number of PCR cycles or sequenced without amplification if possible" to avoid the problem of jumping PCR. Ex. 1015, 9. Kircher,

²⁰ Kircher et al., Double Indexing Overcomes Inaccuracies in Multiplex Sequencing on the Illumina Platform, NUCLEIC ACIDS RESEARCH 40(1):e3 (2011) (Ex. 2004, "Kircher").

²¹ M. Meyer & M. Kircher, *Illumina Sequencing Library Preparation for Highly Multiplexed Target Capture and Sequencing*, COLD SPRING HARBOR PROTOCOL 6 (2010).

however, suggests that “*factors other than jumping PCR* must contribute substantially to the fraction of false pairings.” Ex. 2004, 3 (emphasis added).

Kircher characterizes its “double-indexing method, which places indexes into both of the universal adapter sequences” as “new.” *Id.* at 2. Kircher states that its double-indexing method is a powerful strategy “to improve the accuracy of multiplex genotyping on the Illumina platform.” *Id.* at 7. On this record, thus, we are persuaded by Patent Owner’s argument that Kircher, published months after the priority date of the ’852 patent, is the earliest evidence of researchers “other than the inventors recognizing the problem of cross-over error on Illumina sequencers.” Sur-reply 14.

On this record, we also are persuaded by Patent Owner’s argument that Illumina was not aware of the cross-over error issue until after the priority date of the ’852 patent. *See* PO Resp. 35 (citing Ex. 2001 ¶¶ 128–129; Exs. 2003, 2006, 2007); Sur-reply 14 (citing Ex. 2004). Patent Owner refers to an Illumina document, “Application Note: Sequencing,” which bears a “2011–2012, 2014” copyright date. Ex. 2006, 2. In this document, Illumina suggested single indexing methods for multiplexed 16S microbial sequencing on its platform. *Id.* at 1. This tends to support Patent Owner’s argument that “Illumina’s own documents at the time [of the ’852 patent’s priority date] touted the use of single indexes with its sequencing systems.” PO Resp. 35.

Patent Owner also refers to an Illumina white paper bearing a 2018 copyright date. Ex. 2003, 4. According to this document, Illumina’s sequencing method on flow cells “has been observed to result in higher levels of index misassignment than traditional bridge amplification.” *Id.* at 1–2. Illumina termed the molecular recombination of indexes as “index

hopping.” *Id.* at 1. Illumina stated that it offers “unique dual indexes,” which “eliminate hopped reads from downstream analysis, as unexpected combinations are assigned as undetermined and removed from the data.” *Id.* at 3. This tends to support Patent Owner’s argument that “[i]t was not until around 2017 that Illumina acknowledged cross-over error on its sequencers, and moved toward UDIs.” PO Resp. 35; *see also* Sur-reply 14 (“EX2003 also remains the earliest evidence of *Illumina* finally acknowledging the potential for crossover error on its NGS machines.”).

Petitioner emphasizes that “EX2003 states that ‘[i]ndex misassignment between multiplexed libraries is a *known* issue that has impacted NGS technologies *from the time sample multiplexing was developed*.’” Reply 10 (quoting Ex. 2003, 1, emphasis added by Petitioner). According to Petitioner, this shows “the recognition of chimera formation as problematic as of December 2010.” *Id.* We are not persuaded by Petitioner’s argument.

As Patent Owner correctly points out, the only citation provided by Illumina for the claim that cross-over error was a known issue is to Exhibit 2004, the 2011 Kircher publication. *See* Ex. 2003, 1 (citing reference 2), 4 (showing reference 2 is the Kircher reference (Ex. 2004)). Thus, we agree with Patent Owner that this evidence “further stress[es] that neither Illumina nor researchers were aware of the issue [of cross-over error] at the earlier time of the ’852 Patent invention.” *See* Sur-reply 14.

On this record, we are persuaded by Patent Owner’s argument that “[n]one of Parameswaran, Gloor, or the prevailing views in the art at the time of the invention, identified *template cross-over error* as a source of error that Parameswaran’s indexes could address or reduce at the time of the

patented invention.” PO Resp. 40. We agree with Patent Owner that “[h]aving no awareness of cross-over error, there was no way for a POSA to have any reasonable expectation of success to reduce it.” Sur-reply 20–21.

c. Summary

In sum, on this record, we find Petitioner has not shown sufficiently that the asserted references teach the discarding step. Petitioner also has not shown sufficiently that an ordinarily skilled artisan would have reasonably expected that the combining Parameswaran and Gloor reduces template cross-over error. Thus, Petitioner has not shown by a preponderance of the evidence that claim 1 would have been obvious over the combination of Parameswaran and Gloor.

Each of claims 2–8 depends, directly or indirectly, from claim 1. Thus, we find Petitioner has not shown by a preponderance of the evidence that claims 2–8 would have been obvious over the combination of Parameswaran and Gloor. *See In re Fine*, 837 F.2d 1071, 1076 (Fed. Cir. 1988) (“Dependent claims are nonobvious under section 103 if the independent claims from which they depend are nonobvious.”).

E. Alleged Obviousness over Parameswaran and Bentley

Petitioner asserts that claims 1–8 would have been obvious over the combination of Parameswaran and Bentley. Pet. 51–61. After reviewing the entire record developed at trial, and as explained below, we determine Petitioner has not shown by a preponderance of the evidence that the challenged claims would have been obvious over the asserted prior art.

1. Bentley

Bentley provides a detailed description of Illumina paired-end sequencing. Ex. 1006. It describes a “massively parallel synthetic

sequencing approach” that involved shearing DNA strands into double-stranded DNA fragments for sequencing and ligating (joining) Illumina-specific adapters to both ends. *Id.* at 1–2, 13–14. Bentley “generated high-density single-molecule arrays of genomic DNA fragments attached to the surface of the reaction chamber (the flow cell) and used isothermal ‘bridging’ amplification to form DNA ‘clusters’ from each fragment.” *Id.* at 1. It describes that these DNA fragments were then denatured into single strands so that Illumina-specific adapters on the ends of the templates annealed to complementary oligonucleotides on the surface of the flow cell. *Id.* at 3.

2. Analysis

We, again, focus our analysis on independent claim 1. In this challenge, Petitioner makes similar arguments as those in the challenge based on the combination of Parameswaran and Gloor, except it relies on Bentley instead of Gloor for teaching “amplifying said nucleic acid templates on a surface of a flow cell thereby to form clusters.” Pet. 52–55.

In response, Patent Owner makes similar counterarguments as those in addressing the obviousness challenge based on the combination of Parameswaran and Gloor, namely, Petitioner fails to show a motivation to combine Parameswaran and Bentley to arrive at the claimed invention with a reasonable expectation of success, and the combination does not teach or suggest the discarding step of the claimed method. PO Resp. 51–53.

For the discarding step, Petitioner argues, in its entirety, that “[a]s detailed above in Section VI.A.7, Parameswaran discloses claim 1[f]. EX1003, ¶206.” Pet. 55. For its analysis on reasonable expectation of success, Petitioner argues that

[a] POSA would have had a reasonable expectation of success in combining *Parameswaran* and *Bentley*. EX1003, ¶¶208-211. A POSA would have recognized that adapting the method taught by *Parameswaran* to use Illumina sequencers would have required merely exchanging the Roche-specific adapters for Illumina-specific adapters, and other minor modifications to the sequencing process. *Supra* §VI.C; EX1003, ¶¶208, 166-170. Further, a POSA would have recognized that Illumina sequencing provided sufficient read lengths to sequence the template nucleic acids disclosed in *Parameswaran*. *Supra* §VI.C; EX1003, ¶¶208, 171. A POSA also would have recognized that *Parameswaran*'s unique dual-barcoding approach would desirably reduce cross-over error on an Illumina sequencer. *Supra* §VI.C; EX1003, ¶¶208, 172.

Id. at 56–57.

For the same reasons explained above, we find Petitioner has not shown sufficiently that *Parameswaran* teaches the discarding step. *See supra* Section II.D.3.a. We also find Petitioner has not shown sufficiently that, at the priority date of the '852 patent, an ordinarily skilled artisan would have appreciated cross-over errors resulted from chimeric molecules on Illumina's platform. *See supra* Section II.D.3.b. Because the claimed invention is a method for reducing template cross-over error, Petitioner has not met its burden to show the likelihood of success in combining *Parameswaran* and *Bentley* "to meet the limitations of the claimed invention." *See Intelligent Bio-Sys.*, 821 F.3d at 1367.

As a result, Petitioner has not shown by a preponderance of the evidence that claims 1–8 would have been obvious over the combination of *Parameswaran* and *Bentley*.

III. CONCLUSION

After reviewing the entire record and weighing evidence offered by both parties, we determine that Petitioner has not met its burden to show that

the asserted prior art teaches or suggests the discarding step. Petitioner also has not met its burden to show that an ordinarily skilled artisan would have had a reasonable expectation of success in combining the asserted prior art to arrive at the claimed method. Therefore, Petitioner has not shown, by a preponderance of the evidence, that claims 1–8 would have been obvious over the asserted prior art.

In summary:

Claims	35 U.S.C. §	References	Claims Shown Unpatentable	Claims Not Shown Unpatentable
1–8	103	Parameswaran, Gloor		1–8
1–8	103	Parameswaran, Bentley		1–8
Overall Outcome				1–8

IV. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that Petitioner has not proven, by a preponderance of the evidence, that claims 1–8 of the '852 patent are unpatentable; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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